



Swift System goes open source

Apple's latest intuitive programming language goes open source, and adds Linux support.
<http://dgit.in/nov20-69>

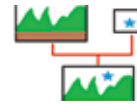


Image Automation

Tutorial on automating Photoshop using Python, to merge images.
<http://dgit.in/nov20-70>



more than 100 countries. The access is free for a few months or if you're on Wi-Fi then you can access it for longer provided your project is limited to a few units. That's what the Starter tier entails, after which you have to pay up to get access to different feature sets of the Particle Cloud service.

Particle's hardware portfolio consists of three boards - Photon, Argon and Boron. Argon is their Wi-Fi line up and Boron are their GSM lineup wherein each board comes with SIM cards that enable you to access the cloud service. The Photon dev board is a Wi-Fi board powered by an STM32 microcontroller and has plenty of I/O ports and switches.

Particle has been on a funding run and has raised a lot of capital over the last year. Moreover, it boasts of a wide client list that includes MIT, Stanford, SpaceX and many others.

THINGWORX

ThingWorx by PTC is an industrial IoT platform that's meant for large-scale IoT applications where you want a robust system that's capable of providing high uptime, quick scalability, tiered-level visibility and control over granular aspects.

It's what you'd opt for if you owned a massive logistics operation spanning multiple factories separated by large geographies. You can even implement it on a smaller scale within a single factory or com-



pany but it's features really shine when you scale up. The features and functionalities of ThingWorx are designed for PLM (Product Lifecycle Management) and it's not meant for small hobbyists wanting to set up a home automation project.

GOOGLE CLOUD IOT CORE

Another major IoT service, Google Cloud IoT Core is a fully-managed service that allows you to manage a

massive network of IoT devices in a secure manner and pull data from all endpoints. All of that data can then be pushed into one of Google's many other tools within GCP to perform various operations. You could move all your data using Google Dataflow into Google BigQuery and perform big data analysis of petabytes of data and even apply machine learning to gain valuable insights. And it's not just that, GCP (Google Cloud Platform) has a variety of tools designed for Big Data ingesting, processing and analysing that allow you to manipulate and examine all that data from your IoT network.

And if you're considering Android Things for your IoT stack then you can connect your project to Cloud IoT Core with just a few lines and get started.

AWS IOT CORE

As per a 2019 Canalsys report, AWS IoT Core enjoys the lion's share of the IoT Cloud market share. Just like Google's offering, AWS IoT Core is also a managed cloud service which allows you to deploy and manage a massive IoT cluster with ease. And you can use HTTP, MQTT and lightweight communication protocol to talk to the IoT devices and pull data from all of them. Since AWS has a bevy of services for big data, you can

